Boolean Connectives and Formal Proofs

Introduction to Logic (CS), Week 2

University of Groningen

Mon 7 Sep - Tue 8 Sep 2026

Today's Lecture

1. Overview
2. Formal Proofs for Conjunction and Disjunction
3. Formal Proofs for Negation and Falsity
4. Conditionals
5. Proofs with conditionals
6. Proofs with (bi)conditionals

Summary of Last Week / Going forward

Propositional (PL) and first-order logic (FOL)

- Atomic sentences ✓
- *Boolean connectives* (this week)
- *Conditionals* (this week)
- Quantifiers

Last week:

- What is logic?
- Valid and sound arguments
- Atomic and compound sentences

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Propositional (PL) and first-order logic (FOL)

- Atomic sentences ✓
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- *Conditionals* (this week)
- Quantifiers

Last week:

- What is logic?
- Valid and sound arguments
- Atomic and compound sentences

This week:

- Boolean connectives: conjunction/disjunction/negation
- Proofs with Boolean connectives
- Strategies and tactics for constructing proofs. Practice is essential!
- Conditionals and hypothetical reasoning

Learning goals

At the end of the week, you will be able to:

- Construct and evaluate compound sentences in PL/FOL using Boolean connectives
- Translation from English to PL/FOL using “and”, “or”, “neither”, etc...
- Construct formal proofs (in $\mathcal{F}$), including nested subproofs, of sentences with Boolean connectives
- Understanding the role of conditionals in logic

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To show validity of an argument, give a proof!

From Language, Proof and Logic p. 46-47:

> *A proof is a step-by-step demonstration that a conclusion follows from some premises...*

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From Language, Proof and Logic p. 46-47:

> *A proof is a step-by-step demonstration that a conclusion follows from some premises...*

1.	P_1	(Premise)
2.	P_2	(Premise)
3.	P_3	(Premise)
4.	T_1	Justification
5.	T_2	Justification
⋮		⋮
k.	C	Justification

Proof rules in $\mathcal{F}$

Which proofs rules have we seen so far?

Proof rules in $\mathcal{F}$

Which proofs rules have we seen so far?

Which rules will we see today?

Proofs with conjunction

Introduction: in order to prove $P \wedge Q$ we have to ...
separately prove P and separately prove Q

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P and we can also conclude Q

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P? Q?

Disjunction elimination

Claim: Given $P \vee Q$, we want to derive R
Proof method (proof by case distinction): Show separately that given P you can derive R, and that given Q you can derive R

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Proof: Consider the number $\sqrt{2}^{\sqrt{2}}$. It is either rational or irrational.

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In case $\sqrt{2}^{\sqrt{2}}$ is rational, we are done: take $b = c = \sqrt{2}$.
In case $\sqrt{2}^{\sqrt{2}}$ is irrational, take $b = \sqrt{2}^{\sqrt{2}}$ and $c = \sqrt{2}$; then

$$(\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^{\sqrt{2}\sqrt{2}} = \sqrt{2}^{2} = 2$$

Disjunction elimination

Claim: n is an integer, then $2n^2 + n + 1$ is not divisible by 3
Proof: Consider three cases: what is the remainder of n divided by 3:

1. Case $n = 3k$
2. Case $n = 3k + 1$
3. Case $n = 3k + 2$

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   - Then $2n^2 + n + 1 = 2(3k + 1)^2 + 3k + 1 + 1 = 3(6k^2 + 5k + 1) + 1$
   - This is not divisible by 3! (Remainder is 1)
3. Case $n = 3k + 2$
   - Then $2n^2 + n + 1 = 2(3k + 2)^2 + 3k + 2 + 1 = 3(6k^2 + 9k + 3) + 2$
   - This is not divisible by 3! (Remainder is 2)

Disjunction elimination

Claim: n is an integer, then $2n^2 + n + 1$ is not divisible by 3

Proof: Consider three cases: what is the remainder of n divided by 3:

1. Case $n = 3k$
   - Then $2n^2 + n + 1 = 2(3k)^2 + 3k + 1 = 3(6k^2 + 1) + 1$
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   - This is not divisible by 3! (Remainder is 2)

In all three cases we have that $2n^2 + n + 1$ is not divisible by 3.

Disjunction elimination

...

...

i. $P \vee Q$

...

| j. P

| ...

| k. R

...

| m. Q

| ...

| n. R

...

s. R $\vee$ Elim: i, j-k, m-n

- Line *i* is the disjunction, subproofs *j*-*k* and *m*-*n* correspond to two cases
- Order of the arguments matters!

Example proofs with disjunction

Try yourself: commutativity and associativity of disjunction

Subproofs

Inside a subproof you may reference an intermediate that is **within the scope**:

1. above it in the same subproof;
2. above it in the main proof or in a subproof of smaller depth

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When a subproof has ended:

1. you may use it in its entirety (for $\neg$Intro or $\vee$Elim, ...)
2. but you **cannot** use individual steps inside the subproof (not even the last step!)

Scoping in programming languages

```
1  void main()
2  {
3    int x = 0;
4    {
5      int y = 2;
6    }
7    {
8      int z = 3;
9      x = 4;
10     y = 0;
11   }
12   z = 1;
13 }
```

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7    {
8      int z = 3;
9      x = 4;
10     y = 0; /* error: use of undeclared identifier 'y' */
11   }
12   z = 1; /* error: use of undeclared identifier 'z' */
13 }
```

A wrong proof

1	$(P \wedge Q) \vee (P \wedge R)$	
2	$\quad P \wedge Q$	
3	$\quad Q$	$\wedge$ Elim: 2
4	$\quad P$	$\wedge$ Elim: 2
5	$\quad P \wedge R$	
6	$\quad P$	$\wedge$ Elim: 5
7	P	$\vee$ Elim: 1, 2–4, 5–6
8	Q	Reit: 3
9	$P \wedge Q$	$\wedge$ Intro: 7, 8

What is wrong with this proof?

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Proof of negation

Claim: $\neg P$

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Proof method: Suppose that P holds, and arrive at a contradiction.

Proof of negation

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Claim: Femke did not eat her sandwitch.
Proof: Suppose that Femke did in fact ate her sandwich. But if we open the fridge, we can see the sandwich right there. “Sandwich was eaten” $\wedge$ “sandwich is in the fridge” is a contradiction. Hence the original assumption that Femke ate her sandwich must be false.

Proof of negation

Claim: $\sqrt{2}$ is not rational (or $\sqrt{2}$ is irrational)
Proof: Suppose that $\sqrt{2}$ is rational.

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$$(\sqrt{2})^2 = 2 = (\frac{a}{b})^2 = \frac{a^2}{b^2}, \quad \text{and} \quad 2b^2 = a^2.$$

Hence a^2 and a are even numbers. Let us write $a = 2c$.

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Hence a^2 and a are even numbers. Let us write $a = 2c$.
Then $2b^2 = (2c)^2 = 4c^2$. Hence b^2 and b are both even. Let us write $b = 2d$. Then the fraction $\frac{a}{b} = \frac{2c}{2d}$ is reducible! Something cannot be both reduced and reducible further, hence we arrived at a contradiction. It means that our original assumption about $\sqrt{2}$ being rational was false. Hence, $\sqrt{2}$ is irrational.

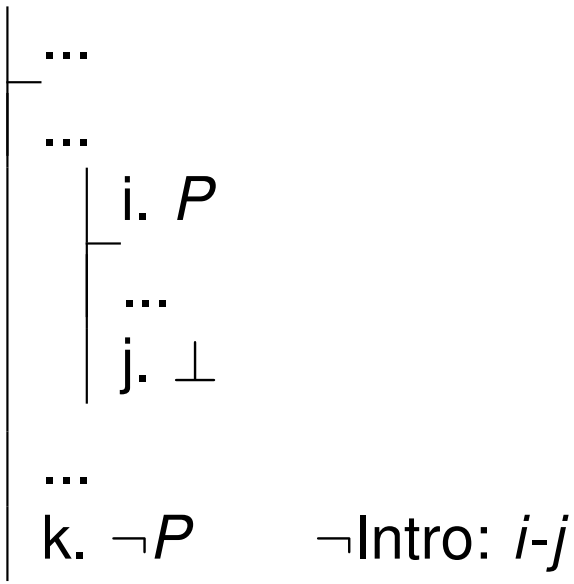
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Proof of negation



- The lines from *i* to *j* denote a *subproof*
- Subproofs can be nested inside each other

Example proof of negation

Not false is true:

$$\vdash \neg\bot$$

From P, deduce P and true:

$$\begin{array}{l} P \\ \vdash \\ P \wedge \neg\bot \end{array}$$

Elimination of negation

Negation has a bit of a special rule for elimination

Double negation elimination:

| ...
|—
| ...
| ...
| i. $\neg\neg P$
| ...
| ...
| j. P $\neg$Elim: i

Contradiction rules

For proving $\neg P$ we need to arrive at a contradiction $\bot$... but how?

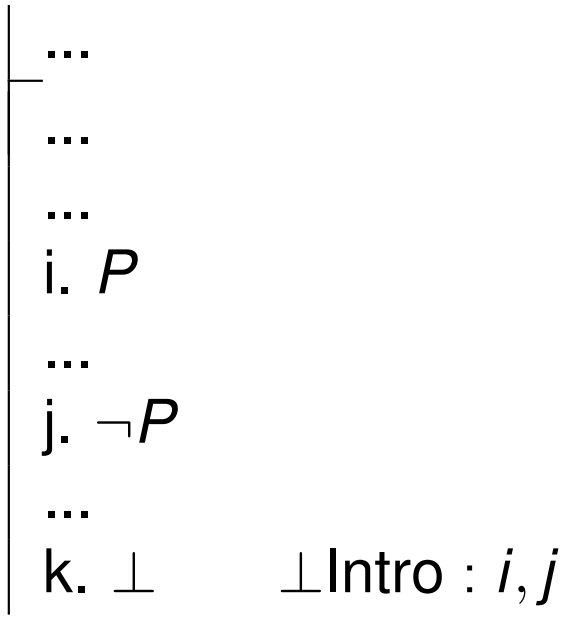
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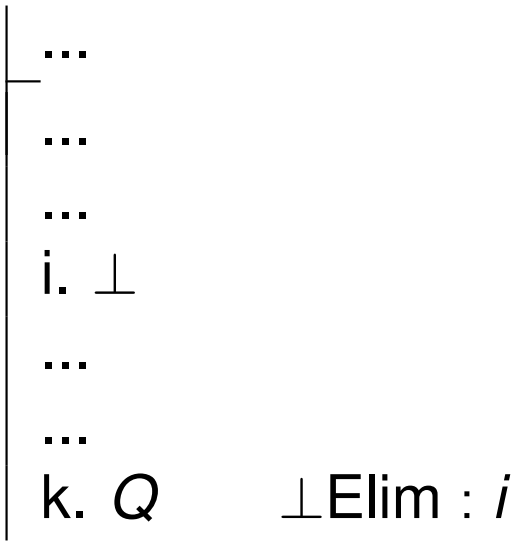
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| ...
|—
| ...
| ...
| i. P
| ...
| j. $\neg P$
| ...
| k. $\bot$ $\bot$Intro : i, j

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Example proofs with $\neg, \bot, \vee$

$$\dfrac{P}{\neg\neg P}$$

$$\dfrac{\begin{array}{l}P \vee Q \\ \neg P\end{array}}{Q}$$

Interesting fact: $\bot$Elim is derivable from $\neg$Intro and $\neg$Elim, i.e. you can prove

$$\dfrac{\bot}{Q}$$

using just the $\neg$ rules.

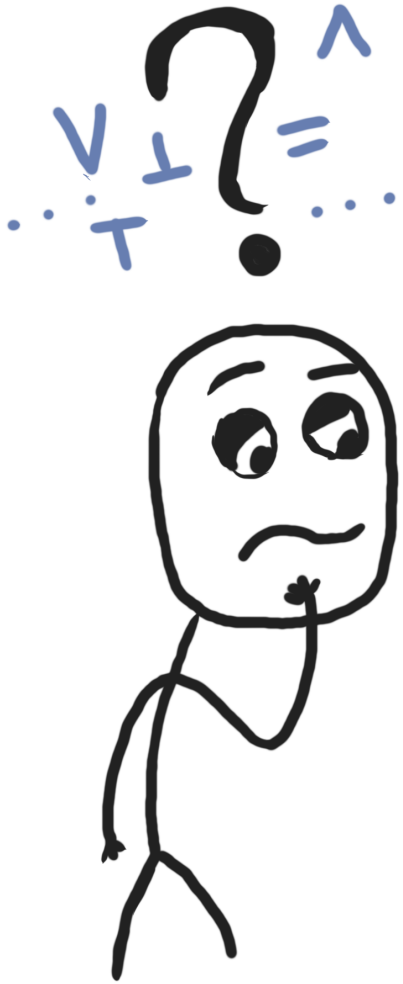
How to find a formal proof?

Like playing chess:

- strict rules,
- tactics and strategies can help

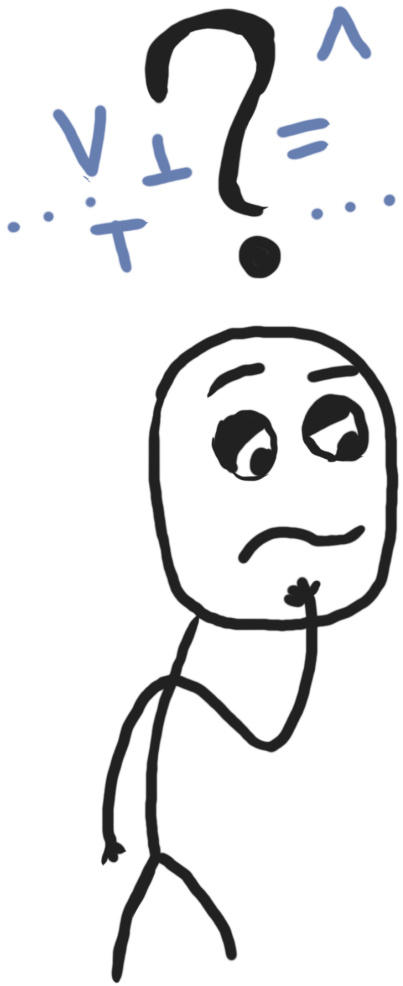


How to approach a proof? (1)



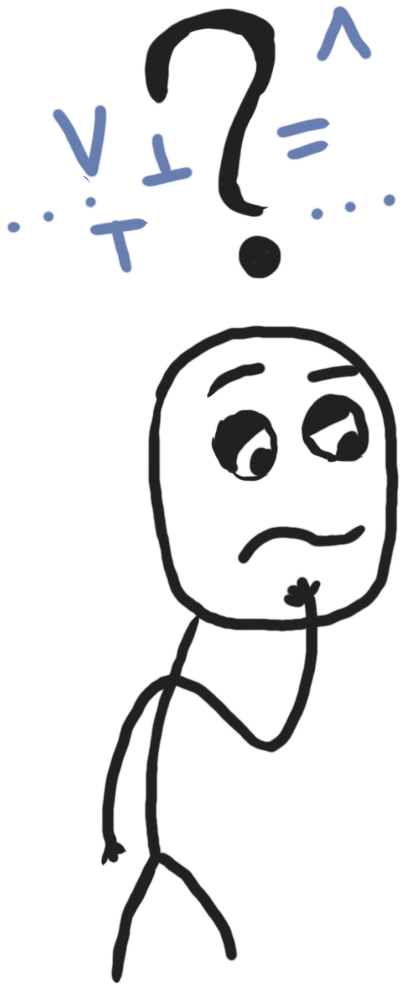
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How to approach a proof? (1)



- Understanding the meaning of the premises and the conclusion
- Use your intuition to verify that the conclusion indeed follows from the premises

How to approach a proof? (1)



- Understanding the meaning of the premises and the conclusion
- Use your intuition to verify that the conclusion indeed follows from the premises
- Try to give an informal argument ("Consider two cases...", "Since we have to prove $\neg P$, we can assume P...")

How to approach a proof? (2)

- Try to work backwards from the conclusion to premises ("To prove $P \wedge Q$ we prove separately P and Q")



How to approach a proof? (2)

- Try to work backwards from the conclusion to premises ("To prove $P \wedge Q$ we prove separately P and Q")
- When working backwards check that your intermediate goals intuitively follow from the premises



How to approach a proof? (2)

- Try to work backwards from the conclusion to premises ("To prove $P \wedge Q$ we prove separately P and Q")
- When working backwards check that your intermediate goals intuitively follow from the premises
- When a direct proof of P is impossible, try proving $\neg\neg P$



Example: ELM

Law of excluded middle

$$\vdash P \vee \neg P$$

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If . . . , then . . .

If you are a member of Cover, then you get a discount.

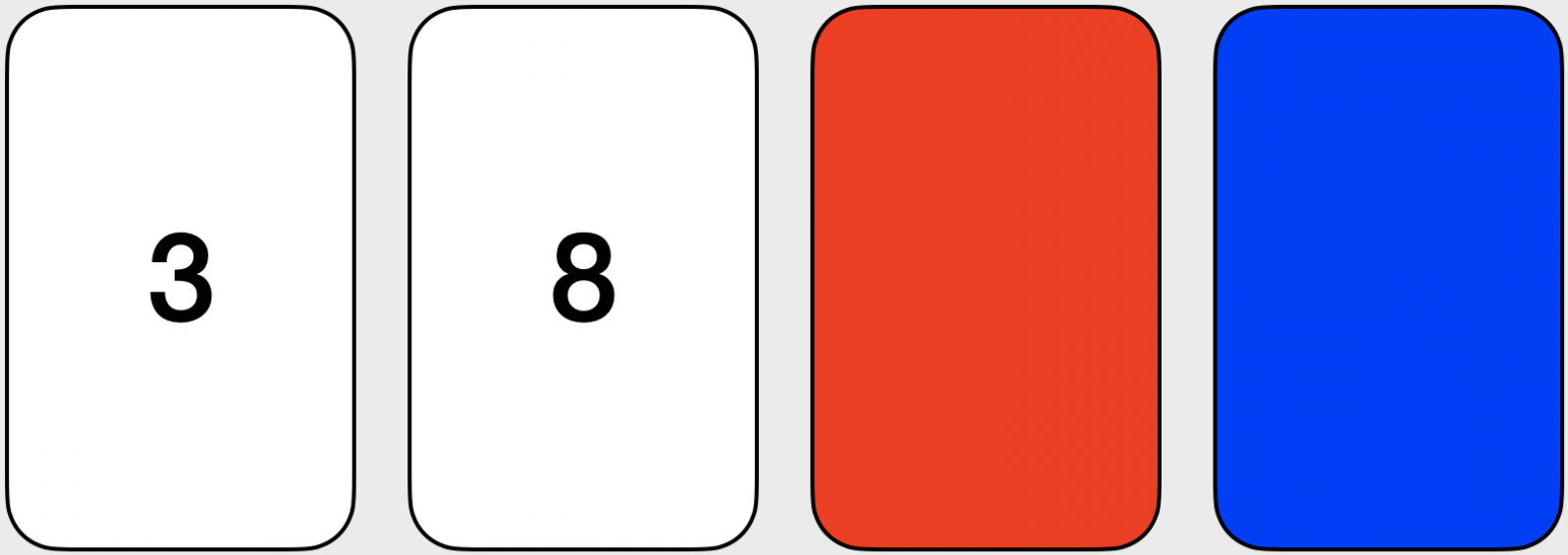
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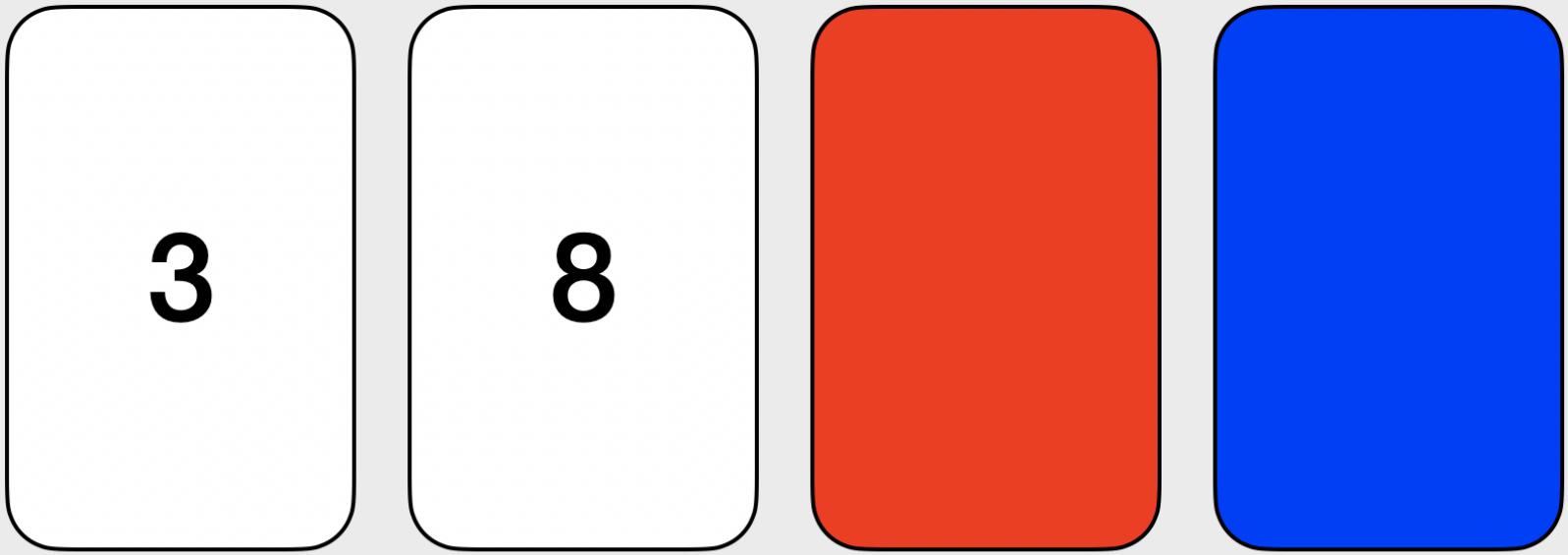
This sentence is a **conditional**.

If $\underbrace{\textit{you are a member of Cover}}_{\text{antecedent}}$, then $\underbrace{\textit{you get a discount}}_{\text{consequent}}$.

Situation: Each card has a number on one side, and a color on the other side.



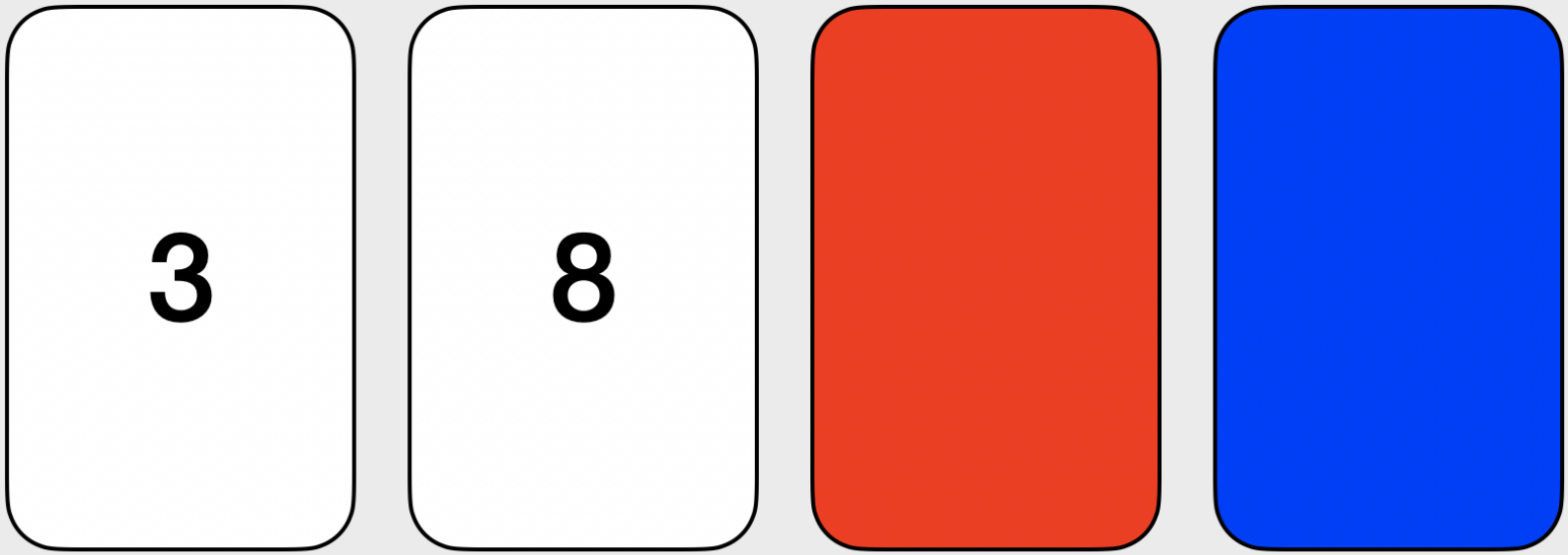
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Proposition:

If a card shows an even number on one side,
then its opposite side is red.

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Question: Which card(s) do you need to turn in order to check whether the proposition is true in this situation?

Situation: Each card represents a person in a bar. The person's age is on one side, and their drink is on the other side.



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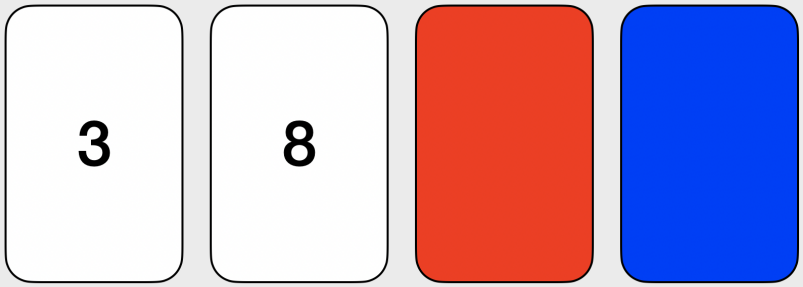
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Question:

Which card(s) do I need to turn in order to check whether the proposition is true in this situation?

Answers

Situation: Each card has a number on one side, and a color on the other side.



Proposition: *If a card shows an even number on one side, then its opposite side is red.*

Cards to turn: 8, Blue

Situation: Each card represents a person in a bar. The person's age is on one side, and their drink is on the other side.



Proposition: *If you are drinking beer, then you must be over 18 years of age.*

Cards to turn: 15, Beer

Human deductive reasoning

We observed that:

- The two tasks were asking the same question: in which situations is a conditional statement true or false.
- In experiments, many more participants got it right in the second task than in the first one.

What do we learn from this?

-
- Conditionals are sometimes difficult understand.
- Depending on context: easy or hard.

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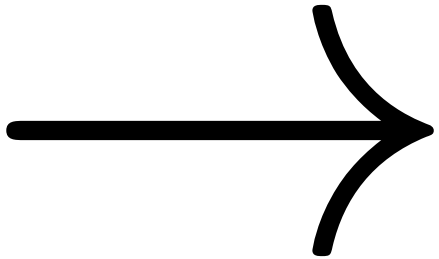
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Conclusion: Rely on the rules of logic, not your intuition.

Conditional: Notation, semantics

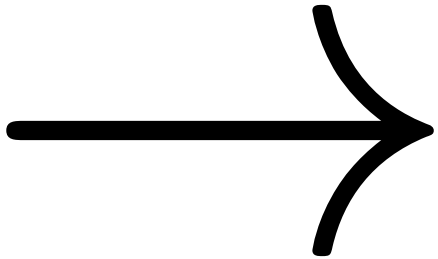


- The sentence $P \to Q$ is a material conditional (or implication);
- We say: "P implies Q"

Truth table:

P	Q	$P \to Q$
T	T	T
T	F	F
F	T	T
F	F	T

Conditional: Notation, semantics



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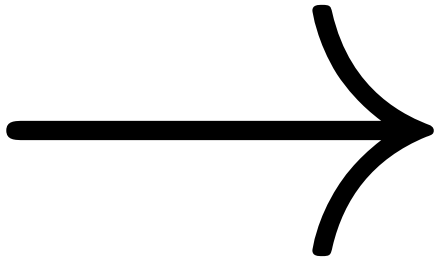
Truth table:

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Compare:

R	S	$R \vee S$
T	T	T
T	F	T
F	T	T
F	F	F

Conditional: Notation, semantics



- The sentence $P \to Q$ is a material conditional (or implication);
- We say: "P implies Q"
- Observe: $P \to Q$ has the same truth table as $\neg P \vee Q$. We write $P \to Q \equiv \neg P \vee Q$.

Truth table:

P	Q	$P \to Q$
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T	F	F
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Compare:

R	S	$R \vee S$
T	T	T
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Necessary and sufficient condition

$$P \to Q$$

says that

- P is a sufficient condition for Q and
- Q is a necessary condition for P.

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For example,

- Being a member of Cover is a *sufficient* condition to get a discount. (But there could be other ways of getting a discount.)
- Going to an exam is a *necessary* condition for passing the course. (But some students can fail the exam.)

Variations of “If . . . , then . . . ”

- The program will crash if you upload a large file.
- The salad will be edible until Sunday, provided that it is refrigerated.
- You are entitled to vote only if you are at least 18.
- Only if you are at least 18, you are entitled to vote.

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P if Q	corresponds to	If Q then P
P provided that Q	corresponds to	If Q then P
Q only if P	corresponds to	If Q then P
Only if P, Q	corresponds to	If Q then P

Logical translation of “unless”

You will miss the train unless you start packing now.

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You will miss the train unless you start packing now.

“*P* unless *Q*” corresponds to “*P* if not *Q*”:

You will miss the train, if you don’t start packing now.

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“*P* unless *Q*” corresponds to “*P* if not *Q*”:

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i.e. “*P* unless *Q*” corresponds to $\neg Q \rightarrow P$.

Logical translation of “unless”

You will miss the train unless you start packing now.

“P unless Q” corresponds to “P if not Q”:

You will miss the train, if you don’t start packing now.

i.e. “P unless Q” corresponds to $\neg Q \rightarrow P$.

But $\neg Q \rightarrow P \equiv \neg\neg Q \vee P \equiv Q \vee P$ (recall that $A \rightarrow B \equiv \neg A \vee B$).

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So “P unless Q” indeed corresponds to $P \vee Q$.

Material Conditional vs Causality

True or false:

- If Groningen is in the Netherlands, then 2+2 = 4.
- If the moon is made of green cheese, then 2+2 = 5.

?

Material Conditional vs Causality

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Both are true according to the truth table semantics of $P \to Q$, but sound rather strange is English.

Main points:

-
- $P \to Q$ carries no causal meaning. It says, if P is true then Q is true.
- $P \to Q$ is true whenever P is false (since $P \to Q \equiv \neg P \vee Q$).

“because” is not truth-functional



Sifan Hassan won gold on the 1500m and 10,000m at the 2019 Athletics World Championships.

-
- Sifan Hassan won the 1500 meters because she was the fastest.

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- Sifan Hassan won the 1500 meters because $2 + 2 = 4$.
  ??? not true.

However, *Sifan Hassan was the fastest* and $2 + 2 = 4$ have the same truth value!

So “because” cannot have a truth-functional interpretation: you cannot define “because” with a truth table.

Condtionals vs if-then-else (1)

If you are a member of Cover, then you get two free drinks, otherwise you get only one free drink.

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> *If you are a member of Cover, then you get two free drinks, otherwise you get only one free drink.*

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General form:

> *If P then Q otherwise R*

$$(P \to Q) \wedge (\neg P \to R)$$

Condtionals vs if-then-else (2)

General form:

If P then Q otherwise R

$$(P \to Q) \wedge (\neg P \to R)$$

Important: This does not directly correspond to `if b then c1 else c2` in programming because `c1` and `c2` are *commands* and not *propositions*.

. . . if and only if . . .

You pass the course if and only if your final grade is at least 6.

. . . if and only if . . .

You pass the course if and only if your final grade is at least 6.

This sentence is a **biconditional** of the sentences

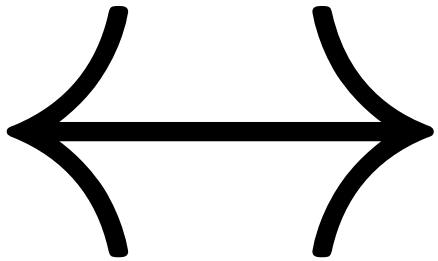
You pass the course

and

your final grade is at least 6.

A grade of at least 6 is a necessary and sufficient condition for passing the course.

Biconditional: Notation, semantics

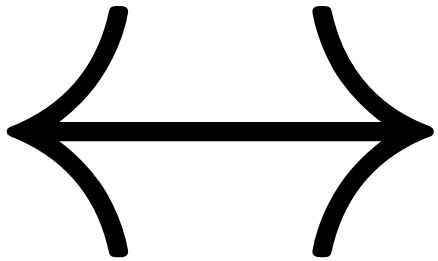


- The sentence $P \leftrightarrow Q$ is a *biconditional*
- We say: "P if and only if Q"

Truth table

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

Biconditional: Notation, semantics



- The sentence $P \leftrightarrow Q$ is a *biconditional*
- We say: "P if and only if Q"
- $P \leftrightarrow Q$ has the same truth table as $(P \rightarrow Q) \wedge (Q \rightarrow P)$

Truth table

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

Variations of “if and only if”

- A number is odd exactly if it is not even.
- n is even iff n^2 is even.
- In a triangle, all sides are equal when all angles are equal.

Logical symbols vs. English words

- $\neg$ corresponds well with not
- $\wedge$ corresponds well with and
- $\vee$ corresponds rather well with or, but *only in the inclusive reading*
- $\rightarrow$ corresponds partially with if . . . then, but *does not cover any causal meaning*
- $\leftrightarrow$ corresponds partially with if and only if, but *does not cover any causal meaning*

Conclusion

- What did we cover today?

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- Homework will be out tomorrow evening-ish, you will have a week to submit it
- Next week: truth tables, equational reasoning